SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct: (192122600891)

I Maithreyan-P *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student Mwy.

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%	2	2	2	2
Debugging	25%	2	2	1	1
Optimization	20%	1	1	1	1
Analysis	20%	1	1	1	1
Code Quality	15%	1	1	1	1

26/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

   - **IP Address:** 192.168.10.65
   - **Subnet Mask:** 255.255.255.192 (/26)
   - **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gate
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

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- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

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NETWORK DEBUGGING AND OPTIMISATION

Assignment Solutions

Note: All text in this document is formatted in black ink. Question 4 repeats Questio
analysis applies.

1. Debugging IPv4 Subnet Configuration

Given: IP address 192.168.10.65, subnet mask 255.255.255.192 (/26), and default

Subnet calculation

- A /26 mask has a block size of 64 addresses in the last octet.
  The /26 subnets in 192.168.10.0/24 begin at .0, .64, .128, and .192.
- The configured address 192.168.10.65 belongs to subnet 192.168.10.64/26.

Item	Value
Network address	192.168.10.64
Broadcast address	192.168.10.127
Valid host range	192.168.10.65 – 192.168.10.126
Usable host addresses	62

Validation and error

- The IP address 192.168.10.65 is valid for 192.168.10.64/26.
- The gateway 192.168.10.1 belongs to 192.168.10.0/26, not 192.168.10.64/26. A host and its normal default gateway must be on the same local subnet.
- Therefore, the workstation is configured for the wrong subnet if it is intended to communicate as part of 192.168.10.0/26.

Corrected configuration

Setting	Correct value
IP address	192.168.10.10
Subnet mask	255.255.255.192 (/26)
Default gateway	192.168.10.1
Network	192.168.10.0/26
Broadcast	192.168.10.63
Valid host range	192.168.10.1 – 192.168.10.62
Usable host addresses	2^(32−26) − 2 = 62

Conclusion: Assigning an unused address such as 192.168.10.10 and keeping gateway 192.168.10.1 places the workstation in the intended subnet. Confirm that the chosen address is not already in use.

2. Debugging IPv6 Address Configuration

Given: Device A = 2001:db8::ff00:42:8329/64; Device B = 2001:db8:0:0:1::1/64.

Full notation and prefixes

Device	Expanded IPv6 address	Network prefix (/64)
A	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:0db8:0000:0000::/64
B	2001:0db8:0000:0000:0001:0000	2001:0db8:0000:0000::/64

	:0000:0001	

Important: A /64 prefix consists of the first four 16-bit groups (the first 64 bits). Device B's fifth group, 0001, is part of its interface identifier, not its /64 network prefix.

Diagnosis

- Both devices belong to the same subnet: 2001:db8:0:0::/64.
- There is no IPv6 subnet-prefix mismatch in the supplied addresses. The premise that the addresses are on different subnets is incorrect.
- Because the addresses are distinct and syntactically valid, check other causes: link/VLAN mismatch, interface status, duplicate addresses, incorrect route or gateway, firewall/ICMPv6 filtering, or neighbor discovery issues.

Optimized configuration and host count

Keep Device A as 2001:db8::ff00:42:8329/64.

- Keep Device B as 2001:db8:0:0:1::1/64; it is already in the same /64 prefix.
- A /64 leaves 64 bits for interface identifiers: 2^64 = 18,446,744,073,709,551,616 possible interface-ID values. Practical usable assignments depend on IPv6 rules and network policy.

Conclusion: Do not change the prefix based on the information given. Troubleshoot Layer 2 connectivity, routes, neighbor discovery, and filtering.

3. IPv4 Subnet Allocation and VLSM Optimisation

Parent network: 192.168.1.0/24. The prompt does not provide each department's required number of hosts or actual assigned addresses. Therefore, exact unused-address counts for each department cannot be determined. The following is a valid example allocation using one /26, one /27, and one /28 without overlap.

Subnet	Network address	Broadcast address	Valid host range	Usable hosts
/26	192.168.1.0	192.168.1.63	192.168.1.1 – 192.168.1.62	62
/27	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94	30
/28	192.168.1.96	192.168.1.111	192.168.1.97 – 192.168.1.110	14

Utilisation analysis

- The three example subnets consume 64 + 32 + 16 = 112 total addresses from the /24 block.
- They provide 62 + 30 + 14 = 106 usable host addresses in total.
- The remaining address space is 192.168.1.112 through 192.168.1.255, which contains 144 addresses that can be allocated to other subnets (subject to correct subnet boundaries).
- Unused host addresses inside a department's subnet = usable host capacity − the number of addresses actually needed/used. The prompt does not state those host counts, so a precise per-department unused count would be a guess.

Common allocation mistakes to check

- Overlapping subnets, such as assigning a /27 that begins inside an already allocated /26.
- Using a network or broadcast address as a host address.
  Assigning a subnet that is too small for the department's host requirement.
- Giving a large subnet to a small department while a larger department runs out of addresses.

Recommended VLSM method

1. List the number of hosts required by every department, including expected growth.
2. Sort departments from the largest host requirement to the smallest.
3. Choose the smallest subnet that supports each requirement. Usable IPv4 hosts are 2^(32−prefix) − 2; for example, /26 supports 62, /27 supports 30, and /28 supports 14.
4. Allocate each subnet on a correctly aligned boundary, without overlap, and record its network, broadcast, and usable range.
5. Reserve some address space for growth and infrastructure, and update the IP address management records.

   Example recommendation: use /26 for a department needing up to 62 hosts, /27 for up to 30, and /28 for up to 14. If actual requirements differ, recalculate the VLSM plan rather than assigning masks solely by department name.

   Conclusion: VLSM improves utilisation by matching subnet size to host demand. A final optimised plan requires each department's host count.

4. IPv6 Configuration (Repeated Question)

Question 4 is identical to Question 2. The expanded addresses are:

- Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329/64
- Device B: 2001:0db8:0000:0000:0001:0000:0000:0001/64
- Both share the network prefix 2001:0db8:0000:0000::/64, so they are in the same /64 subnet.
- No prefix correction is needed. Check VLAN/LAN connectivity, interface status, routes, duplicate addresses, neighbor discovery, and firewall rules.
- A /64 provides 2^64 possible 64-bit interface identifiers.

5. Debugging the Routing Table

Destination network	Next hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route (0.0.0.0/0)	10.0.0.1

Longest-prefix match for 192.168.2.100

- 192.168.2.100 falls within 192.168.2.0/24 because the first 24 bits match.
- The most specific matching entry is 192.168.2.0/24, so the router selects next hop 10.0.0.3.
- The default route is not selected when a more specific matching route exists.

Possible errors and debugging steps

- The table itself contains the appropriate destination route for 192.168.2.100. If forwarding fails, verify that 10.0.0.3 is reachable from this router and that it has a valid route to 192.168.2.0/24.
- Check the destination network's subnet mask, interface status, return route, access-control lists/firewall rules, and whether the next-hop address is correct.
- Use routing-table inspection and ping/traceroute tests to identify where packets stop. Do not change the next hop without verifying the actual topology.

Optimized routing table

Keep the two specific /24 routes and the default route if they represent the real network topology. Ensure each next hop is reachable on a directly connected link, remove obsolete or duplicate routes, and use route summarisation only if it accurately represents reachable networks.

Destination	Selected next hop	Reason
192.168.2.100	10.0.0.3	Matches 192.168.2.0/24 (longest prefix)
An address in 192.168.1.0/24	10.0.0.2	Matches 192.168.1.0/24
Destination with no more-specific matching route	10.0.0.1	Uses the default route 0.0.0.0/0

Conclusion: The correct longest-prefix match for 192.168.2.100 is 192.168.2.0/24 via 10.0.0.3. If delivery fails, investigate next-hop reachability and the onward/return paths.

Overall Conclusion

The IPv4 problem is caused by a workstation address and gateway belonging to different /26 subnets. The IPv6 addresses actually share the same /64 prefix, so the failure must be investigated elsewhere. The VLSM question needs host requirements for exact unused-address calculations; a valid non-overlapping example and optimisation method are provided. In the routing problem, longest-prefix match selects 192.168.2.0/24 via 10.0.0.3, while unmatched destinations use the default route via 10.0.0.1.